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Project Title: Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator

Week: Week 5

WEEK-5 TASK

Task Name

Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator

Abstract

A Dynamic Portfolio Risk Arbitrage and Value-at-Risk (VaR) Monte Carlo Simulator is a Business Analytics model used to measure the risk of a portfolio containing multiple assets. The model generates sample historical prices, calculates asset returns and the historical covariance matrix, and applies Cholesky decomposition to create correlated random movements. Geometric Brownian Motion (GBM) is then used to simulate future asset prices. The simulated portfolio returns are analyzed to estimate the portfolio's 99% Value-at-Risk (VaR).

Introduction

Portfolio risk analysis is important because investors need to understand how much money they could potentially lose under unfavorable market conditions. Business Analytics helps convert historical market data into useful risk measures. Monte Carlo simulation allows many possible future price movements to be generated, while covariance and Cholesky decomposition preserve relationships between assets. The resulting 99% VaR gives an estimate of the potential portfolio loss at a very high confidence level.

Objectives

1. To generate and analyze historical price data for multiple assets.
2. To calculate daily returns and the historical covariance matrix.
3. To use Cholesky decomposition to correlate multi-dimensional random noise.
4. To simulate future asset prices using Geometric Brownian Motion.
5. To calculate portfolio profit and loss from simulated outcomes.

6. To estimate the portfolio's 99% Value-at-Risk using Monte Carlo simulation.
7. To support data-driven portfolio risk management.

Monte Carlo VaR Model

Input Data

The simulator uses a sample portfolio containing multiple assets. For a simple implementation, historical daily prices are generated for 500 assets. The model also uses asset weights, initial prices, expected returns, volatility, simulation period, and the number of Monte Carlo simulations.

Covariance Matrix

Daily returns are calculated from historical prices. The covariance matrix measures how asset returns move together. A positive covariance indicates that two assets tend to move in the same direction, while a negative covariance indicates opposite movement.

Cholesky Decomposition

Cholesky decomposition converts the covariance matrix into a lower triangular matrix. This matrix is multiplied by independent random numbers to produce correlated random movements that represent the relationships between assets.

Geometric Brownian Motion

Geometric Brownian Motion (GBM) is used to simulate future asset prices. The model uses the initial price, expected return, volatility, time period, and correlated random noise to generate possible future prices.

Value-at-Risk (VaR)

Value-at-Risk estimates the potential portfolio loss over a selected time period at a specified confidence level. In this project, the 99% VaR is calculated from the simulated portfolio returns. The 1st percentile of the simulated profit/loss distribution is used to estimate the potential loss.

Simulation Results & Observations

The simulator generates a large number of possible future portfolio returns. The covariance matrix captures the historical relationship between assets, and Cholesky decomposition introduces these relationships into the random simulations. The GBM model produces possible future prices, which are converted into portfolio returns. The

99% VaR represents the estimated loss that should not be exceeded in approximately 99% of the simulated outcomes under the model assumptions.

Risk Analysis

• Higher asset volatility generally increases portfolio risk.

• Strong positive correlations between assets can increase portfolio risk.

• Diversification across less-correlated assets can reduce portfolio risk.

• The 99% VaR provides a high-confidence estimate of potential portfolio loss.

• Monte Carlo simulation can evaluate many possible market scenarios.

Benefits

• Provides a quantitative measure of portfolio risk.

• Considers relationships between multiple assets.

• Generates many possible future market scenarios.

• Supports investment and risk-management decisions.

• Provides a clear 99% VaR measure for management.

Conclusion

The Dynamic Portfolio Risk Arbitrage and Value-at-Risk (VaR) Monte Carlo Simulator demonstrates how Business Analytics can be used for financial risk management. By combining historical returns, covariance analysis, Cholesky decomposition, Geometric Brownian Motion, and Monte Carlo simulation, the model estimates the potential risk of a multi-asset portfolio. The 99% VaR gives management an easy-to-understand measure of potential portfolio loss and supports data-driven investment decisions.